Reality of the common zeros for centrohermitian Toeplitz matrices

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Task 1. Resolve the common-zero reality conjecture

For $m \geq 1$ and $n \geq 0$, let I_k be the $m \times (m+n)$ matrix whose (i,j) entry is 1 if $j - i = k$ and 0 otherwise, with indices starting at zero. For $A \in \mathbb{C}^{m\times(m+n)}$, put

$$B(z) = A + \sum_{k=0}^{n} z_k I_k.$$

Its generalized characteristic polynomials are all $m \times m$ minors of $B(z)$. Their common zeros are exactly the points where $\operatorname{rank} B(z) < m$. A matrix is centrohermitian when $A_{i,j} = \overline{A_{m-1-i,m+n-1-j}}$. We prove Conjecture 3.2 of Xu [1], corresponding to the reality conjecture of Alexandersson and Shapiro [2].

Theorem 1. *If $A \in \mathbb{C}^{m\times(m+n)}$ is Toeplitz and centrohermitian, then every common zero z of its generalized characteristic polynomials satisfies*

$$z_j = \overline{z_{n-j}} \qquad (0 \leq j \leq n).$$

There is no genericity, bandwidth, simplicity, or corank-one assumption.

Proof. Write $A_{i,j} = a_{j-i}$. Centrohermitian symmetry says precisely that

$$a_q = \overline{a_{n-q}} \quad (-(m-1) \leq q \leq m+n-1). \tag{1}$$

Let z be a common zero. Choose a nonzero column vector $v \in \mathbb{C}^m$ such that $v^* B(z) = 0$. For $0 \leq j, k \leq n$, define

$$u_j = I_j^* v, \qquad H_{jk} = v^* I_k I_j^* v = u_k^* u_j, \qquad b_j = v^* A I_j^* v.$$

Multiplying $v^* B(z) = 0$ by $I_j^* v$ gives the linear system

$$Hz = -b. \tag{2}$$

We first check that H is invertible. The vectors $u_0, \ldots, u_n$ are linearly independent: if the first nonzero coordinate of v is v_r, then the submatrix of $[u_0 \ \cdots \ u_n]$ on rows $r, r+1, \ldots, r+n$ is lower triangular with diagonal v_r. All those rows exist because $r+n \leq m+n-1$. Moreover, for $c \in \mathbb{C}^{n+1}$,

$$c^* H c = \left\| \sum_{j=0}^{n} \overline{c_j} u_j \right\|^2 .$$

Thus H is positive definite. The conjugates in this formula account for the fact that H is the transpose of the usual Gram matrix.

Extend v_i by zero outside $0 \le i < m$. Directly from the shift matrices,

$$H_{jk} = \sum_{i \in \mathbb{Z}} \overline{v_i} v_{i+k-j}. \tag{3}$$

Consequently H is Hermitian Toeplitz and

$$H_{jk} = \overline{H_{n-j,n-k}}. \tag{4}$$

The right-hand side has the same conjugate-reversal symmetry. Indeed,

$$b_j = \sum_{i,l=0}^{m-1} \overline{v_i}\, a_{l+j-i}\, v_l,$$

and, by interchanging i, l and using (1),

$$\begin{aligned} \overline{b_{n-j}} &= \sum_{i,l=0}^{m-1} \overline{v_i}\, \overline{a_{i+n-j-l}}\, v_l \\ &= \sum_{i,l=0}^{m-1} \overline{v_i}\, a_{l+j-i}\, v_l = b_j. \end{aligned}$$

Let J reverse the $n+1$ coordinates. Equations (4) and the identity for b say $H = J\overline{H}J$ and $b = J\bar{b}$. Conjugating (2) and multiplying by J therefore gives

$$H(J\overline{z}) = -b.$$

Since H is invertible, (2) has only one solution. Hence $z = J\overline{z}$, as required. □

The proof uses only a nonzero left null vector and finite-dimensional linear algebra. In particular, it applies to every common zero, including multiple zeros and points where the rank drops by more than one. The sign convention $A - \sum z_k I_k$ used in [2] gives the same conclusion after replacing z by $-z$.

Task 2. Quantify stability under structured perturbations

Obtain explicit bounds on the distance from an approximate common zero to $\{z : z = J\overline{z}\}$ in terms of a left-kernel residual and the defect of Toeplitz and centrohermitian symmetry. Determine sharp dimension-dependent lower bounds for the smallest eigenvalue of $H(v)$ when $\|v\| = 1$.

Task 3. Determine the geometry of the real eigenvalue locus

Use $z = -H(v)^{-1}b(v)$ to obtain location bounds and to study the limiting support of the common zeros of growing finite sections. Identify when this parametrization describes the boundary of that support.

Task 4. Characterize simple and multiple common zeros

Find conditions on the Toeplitz coefficients that make all common zeros simple, and relate higher multiplicities to the left-kernel dimension and to the local determinantal equations.

Task 5. Classify matrix-valued extensions

Determine which adjoint symmetries on block Toeplitz coefficients and which choices of parameter matrices preserve an analogous reality theorem. Compare scalar parameters multiplying identity blocks with independent matrix-valued parameters.

References

[1] Y. Xu, *Generalized characteristic polynomials and Gaussian cubature rules*, SIAM J. Matrix Anal. Appl. **36** (2015), 1129–1142. arXiv:1406.3329v2, Conjecture 3.2, manuscript p. 7.

[2] P. Alexandersson and B. Shapiro, *Around a multivariate Schmidt–Spitzer theorem*, Linear Algebra Appl. **446** (2014), 356–368. arXiv:1302.3716; Conjecture 10 in the author-hosted revised manuscript.